# AI - Assisted Coding-Assignment -14.3

**HTNO:2303A52228**
**Batch:37**

**Lab 14: Web Design Applications Using AI Assistance HTML, CSS, and JavaScript Generation with GitHub Copilot.**

**Task Description-1**
AI-Assisted Personal Portfolio Website.
**Expected Output:**
Clean and well-structured portfolio website
Responsive layout for desktop and mobile screens
Interactive project cards
**Prompt:**
Generate a simple personal portfolio website using HTML, CSS, and JavaScript. The webpage should contain sections for Home, About, Projects, and Contact with a clean and responsive design. Use CSS Flexbox or Grid to arrange the layout and add hover effects for project cards. Ensure the webpage looks organized on both desktop and mobile screens and include smooth scrolling navigation.
**CODE AND OUTPUT :**

index.html ● # style.css ●

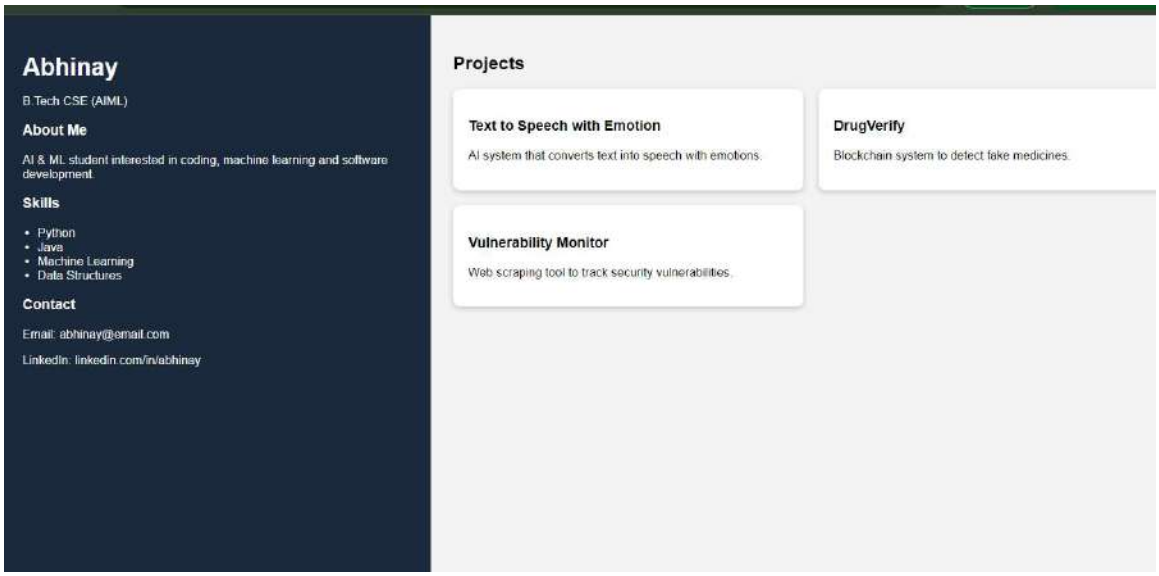
index.html > html > body > footer

```html
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>My Portfolio</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header>
<nav>
<h2 class="logo">MyPortfolio</h2>
<ul class="nav-links">
<li><a href="#home">Home</a></li>
<li><a href="#about">About</a></li>
<li><a href="#projects">Projects</a></li>
<li><a href="#contact">Contact</a></li>
</ul>
</nav>
</header>
<section id="home" class="home">
<h1>Hello, I'm Abhinay</h1>
<p>B.Tech CSE (AIML) Student</p>
</section>
<section id="about" class="about">
<h2>About Me</h2>
<p>
I am a Computer Science student specializing in AI & Machine Learning.
I enjoy solving coding problems and building ML projects.
</p>
```

index.html ● # style.css ●

# style.css > .nav-links li

```
*{
margin:0;
padding:0;
box-sizing:border-box;
font-family:Arial;
}
html{
scroll-behavior:smooth;
}
header{
background: #222;
padding:15px;
}
nav{
display:flex;
justify-content:space-between;
align-items:center;
color: white;
}
.nav-links{
display:flex;
list-style:none;
}
.nav-links li{
margin-left:20px;
}
.nav-links a{
color: white;
text-decoration:none;
font-weight:bold;
```



**Explanation:**
The AI-assisted personal portfolio website was created to display a student's academic information, skills, projects, and contact details in a single webpage. The structure of the website was developed using HTML, while CSS was used to design a responsive and attractive layout. AI coding assistance from GitHub Copilot helped generate the initial code for the webpage structure and styling. The generated code was then customized by adding hover effects on project cards and smooth navigation to improve user experience.

**Task Description-2**
AI-Generated Restaurant Landing Page
**Expected Output:**
Fully functional restaurant landing page
Responsive navigation bar
Working image slider with smooth transitions
**Prompt:**
Generate a responsive restaurant landing page using HTML, CSS, and JavaScript. Include a navigation bar with Home, Menu, Gallery, and Contact links. Create a menu section with CSS card styling. Add a JavaScript image slider for the gallery with smooth transitions.
**CODE AND OUTPUT :**

index.html > ...

```html
<!DOCTYPE html>
<html>
<head>
<title>Restaurant Landing Page</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<!-- Navigation Bar -->
<nav>
<h2 class="logo">Food Corner</h2>
<ul>
<li><a href="#home">Home</a></li>
<li><a href="#menu">Menu</a></li>
<li><a href="#gallery">Gallery</a></li>
<li><a href="#contact">Contact</a></li>
</ul>
</nav>
<!-- Home Section -->
<section id="home">
<h1>Welcome to Food Corner</h1>
<p>Enjoy Delicious Food</p>
</section>
<!-- Menu Section -->
<section id="menu">
<h2>Our Menu</h2>
<div class="menu-container">
<div class="card">
<h3>Pizza</h3>
<p>Cheese Pizza</p>
</div>
```

index.html > ...

```html
<html>
<body>
<section id="menu">
<div class="menu-container">
<div class="card">
<h3>Burger</h3>
<p>Chicken Burger</p>
</div>
<div class="card">
<h3>Pasta</h3>
<p>Italian Pasta</p>
</div>
</div>
</section>
<!-- Gallery Section -->
<section id="gallery">
<h2>Gallery</h2>
<div class="slider">
<img id="slide" src="https://picsum.photos/500/300?1">
</div>
</section>
<!-- Contact Section -->
<section id="contact">
<h2>Contact Us</h2>
<p>Email: foodcorner@gmail.com</p>
<p>Phone: 9876543210</p>
</section>
<script src="script.js"></script>
</body>
</html>
```

style.css > body

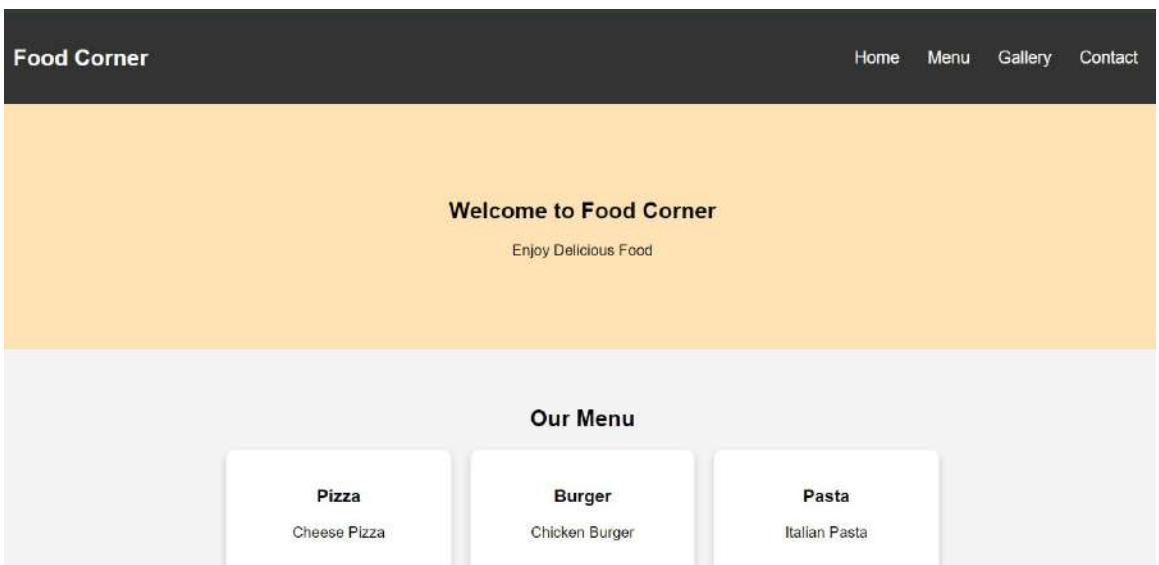
```
body{
margin:0;
font-family:Arial;
background:#f4f4f4;
}
/* Navigation Bar */
nav{
display:flex;
justify-content:space-between;
align-items:center;
background:#333;
padding:15px;
}
.logo{
color:white;
}
nav ul{
list-style:none;
display:flex;
}
nav ul li{
margin:0 15px;
}
nav ul li a{
color:white;
text-decoration:none;
font-size:18px;
transition:0.3s;
}
nav ul li a:hover{
```

# style.css > body

```
nav ul li a:hover{
color: orange;
}
/* Home Section */
#home{
text-align:center;
padding:80px;
background: #ffe4b5;
}
/* Menu Section */
#menu{
text-align:center;
padding:40px;
}
.menu-container{
display:flex;
justify-content:center;
gap:20px;
flex-wrap:wrap;
}
.card{
background: white;
padding:20px;
width:200px;
border-radius:10px;
box-shadow:0 4px 8px rgba(0,0,0,0.2);
transition:0.3s;
}
.card:hover{
transform:scale(1.1);
```

JS script.js > ...

```javascript
var images=[
"https://picsum.photos/500/300?1",
"https://picsum.photos/500/300?2",
"https://picsum.photos/500/300?3"
];
var index=0;
function changeImage(){
index++;
if(index>=images.length){
index=0;
}
document.getElementById("slide").src=images[index];
}
setInterval(changeImage,2000);
```




**Explanation:**
AI tools such as GitHub Copilot were used to quickly generate the HTML structure, CSS styling, and JavaScript functionality. The navigation bar allows users to easily move between sections of the page. Menu items are displayed using card-based layouts for a modern UI design. A JavaScript image slider was implemented to automatically display restaurant images with smooth transitions, improving the user experience.

**Task Description-3**
AI-Powered Event Registration Form.

**Expected Output:**
User-friendly and visually appealing form
Real-time validation messages
Prevention of invalid submissions.
**Prompt:**
Generate a modern event registration form using HTML, CSS, and JavaScript for a university technical fest. Include fields for Name, Email, Phone Number, Department, and Event selection. Add CSS styling for a professional look and implement JavaScript validation for email format, phone number length, and required fields with real-time error messages.
**CODE AND OUTPUT:**

index.html > ...

```html
<!DOCTYPE html>
<html>
<head>
<title>SR University Tech Fest Registration</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<div class="container">
<h1>SR University Tech Fest</h1>
<p class="subtitle">Event Registration Form</p>
<form id="form">
<div class="input-group">
<label>Name</label>
<input type="text" id="name" placeholder="Enter your name">
<span class="error" id="nameError"></span>
</div>
<div class="input-group">
<label>Email</label>
<input type="text" id="email" placeholder="Enter your email">
<span class="error" id="emailError"></span>
</div>
<div class="input-group">
<label>Phone Number</label>
<input type="text" id="phone" placeholder="Enter phone number"
<span class="error" id="phoneError"></span>
</div>
<div class="input-group">
<label>Department</label>
<select id="department">
<option value="">Select Department</option>
```

```html
<option>CSE</option>
<option>ECE</option>
<option>MECH</option>
<option>CIVIL</option>
</select>
<span class="error" id="deptError"></span>
</div>
<div class="input-group">
<label>Event</label>
<select id="event">
<option value="">Select Event</option>
<option>Coding Challenge</option>
<option>Hackathon</option>
<option>Robotics</option>
<option>AI Workshop</option>
</select>
<span class="error" id="eventError"></span>
</div>
<button type="submit">Register</button>
</form>
</div>
<script src="script.js"></script>
</body>
</html>
```

# style.css > body

```
body{
font-family:Arial;
background:linear-gradient(135deg, #4facfe, #00f2fe);
height:100vh;
display:flex;
justify-content:center;
align-items:center;
margin:0;
}
.container{
background: white;
padding:40px;
border-radius:10px;
width:350px;
box-shadow:0 8px 20px rgba(0,0,0,0.2);
}
h1{
text-align:center;
margin-bottom:5px;
}
.subtitle{
text-align:center;
color: gray;
margin-bottom:20px;
}
.input-group{
margin-bottom:15px;
display:flex;
flex-direction:column;
}
```

# style.css > body

```
.input-group{
}
label{
margin-bottom:5px;
font-weight:bold;
}
input,select{
padding:10px;
border:1px solid #ccc;
border-radius:5px;
font-size:14px;
}
input:focus,select:focus{
outline:none;
border-color: #4facfe;
}
button{
width:100%;
padding:12px;
background: #4facfe;
color: white;
border:none;
border-radius:5px;
font-size:16px;
cursor:pointer;
transition:0.3s;
}
button:hover{
background: #0077ff;
}
```

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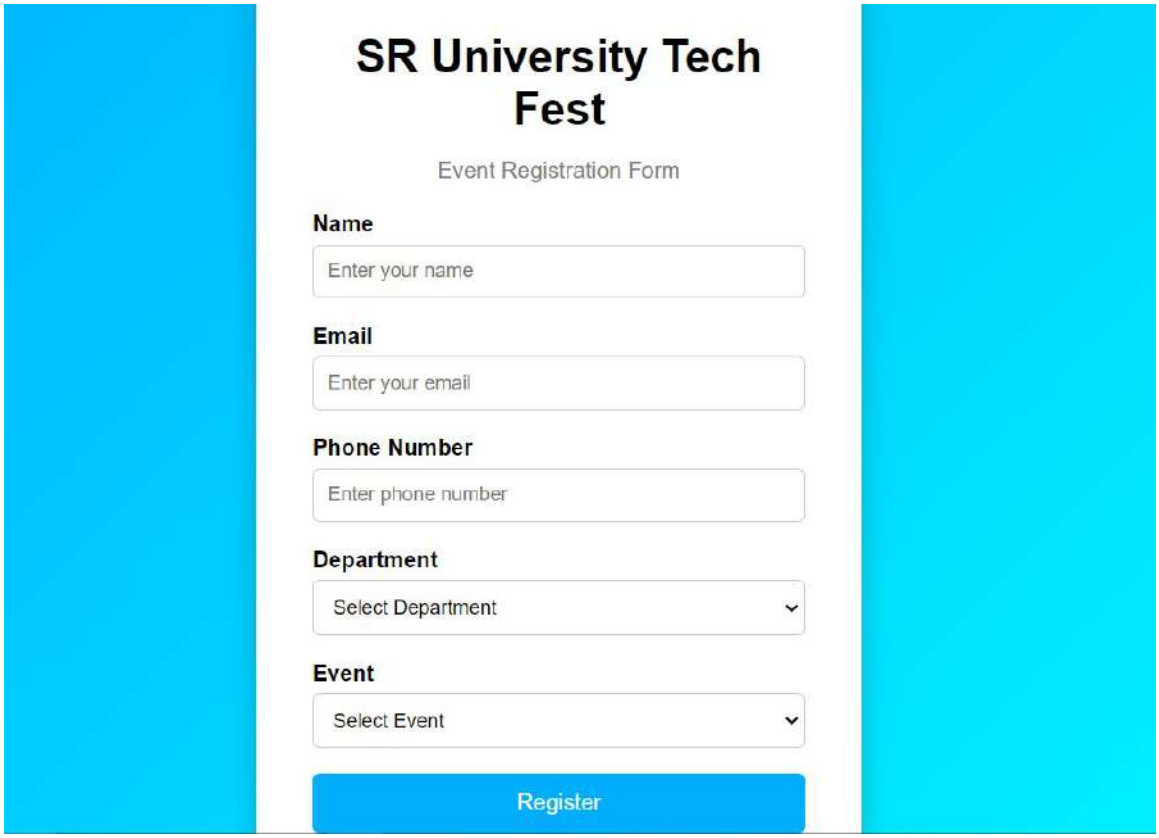
JS script.js > ...

```js
var form=document.getElementById("form");
form.addEventListener("submit",function(e){
var name=document.getElementById("name").value;
var email=document.getElementById("email").value;
var phone=document.getElementById("phone").value;
var dept=document.getElementById("department").value;
var event=document.getElementById("event").value;
var emailPattern=/^[^ ]+@[^ ]+\.[a-z]{2,3}$/;
var valid=true;
/* Name Validation */
if(name==""){
document.getElementById("nameError").innerText="Name is requir
valid=false;
}else{
document.getElementById("nameError").innerText="";
}
/* Email Validation */
if(!email.match(emailPattern)){
document.getElementById("emailError").innerText="Enter valid e
valid=false;
}else{
document.getElementById("emailError").innerText="";
}
/* Phone Validation */
if(phone.length!=10){
document.getElementById("phoneError").innerText="Phone must be
valid=false;
}else{
document.getElementById("phoneError").innerText="";
}
```




**Explanation:**
AI tools like GitHub Copilot were used to assist in developing the event registration form for SR University's technical fest. The HTML code was generated to create a structured form that collects user details such as name, email, phone number, department, and selected event. CSS styling was applied to make the form visually appealing, readable, and user-friendly with proper spacing, colors, and layout.

**Task Description-4**
AI-Assisted E-Commerce Product.
**Expected Output:**
Functional product catalog
Working cart counter
Interactive user experience
**Prompt:**
Generate a simple e-commerce product page using HTML, CSS, and JavaScript. Create a grid-based product catalog with images, product names, and prices. Implement an "Add to Cart" button using JavaScript that updates a cart counter at the top-right corner and provides visual feedback when a product is added.
**CODE AND OUTPUT :**

index.html > html > head

```html
<!DOCTYPE html>
<html>
<head>
<title>Simple E-Commerce Store</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header>
<h2>Tech Store</h2>
<div class="cart">
🛒 Cart <span id="cart-count">0</span>
</div>
</header>
<section class="products">
<div class="product">
<img src="https://picsum.photos/200?1">
<h3>Wireless Headphones</h3>
<p>₹1999</p>
<button onclick="addToCart(this)">Add to Cart</button>
</div>
<div class="product">
<img src="https://picsum.photos/200?2">
<h3>Smart Watch</h3>
<p>₹2999</p>
<button onclick="addToCart(this)">Add to Cart</button>
</div>
<div class="product">
<img src="https://picsum.photos/200?3">
<h3>Bluetooth Speaker</h3>
<p>₹1499</p>
```

```html
<html>
<body>
<section class="products">
<div class="product">
<button onclick="addToCart(this)">Add to Cart</button>
</div>
<div class="product">
<img src="https://picsum.photos/200?3">
<h3>Bluetooth Speaker</h3>
<p>₹1499</p>
<button onclick="addToCart(this)">Add to Cart</button>
</div>
<div class="product">
<img src="https://picsum.photos/200?4">
<h3>Gaming Mouse</h3>
<p>₹999</p>
<button onclick="addToCart(this)">Add to Cart</button>
</div>
</section>
<script src="script.js"></script>
</body>
</html>
```

```
1  body{
2  font-family:Arial;
3  margin:0;
4  background:#f4f4f4;
5  }
6  header{
7  display:flex;
8  justify-content:space-between;
9  align-items:center;
10 padding:20px;
11 background:#333;
12 color:white;
13 }
14 .cart{
15 font-size:18px;
16 }
17 #cart-count{
18 background:red;
19 padding:4px 8px;
20 border-radius:50%;
21 margin-left:5px;
22 }
23 .products{
24 display:grid;
25 grid-template-columns:repeat(auto-fit,minmax(200px,1fr));
26 gap:20px;
27 padding:40px;
28 }
29 .product{
30 background:white;
```

Ln 1, C

style.css > body

```
29  .product{
31  padding:20px;
32  text-align:center;
33  border-radius:10px;
34  box-shadow:0 4px 10px rgba(0,0,0,0.2);
35  transition:0.3s;
36
37
38
39
40
41
42  border-radius:8px;
43  }
44  button{
45  margin-top:10px;
46  padding:10px 15px;
47  border:none;
48  background: #0077ff;
49  color: white;
50  border-radius:5px;
51  cursor:pointer;
52  transition:0.3s;
53  }
54  button:hover{
55  background: #0055cc;
56  }
57  .added{
58  background: green !important;
```

Defines the radii of the outer border edge.

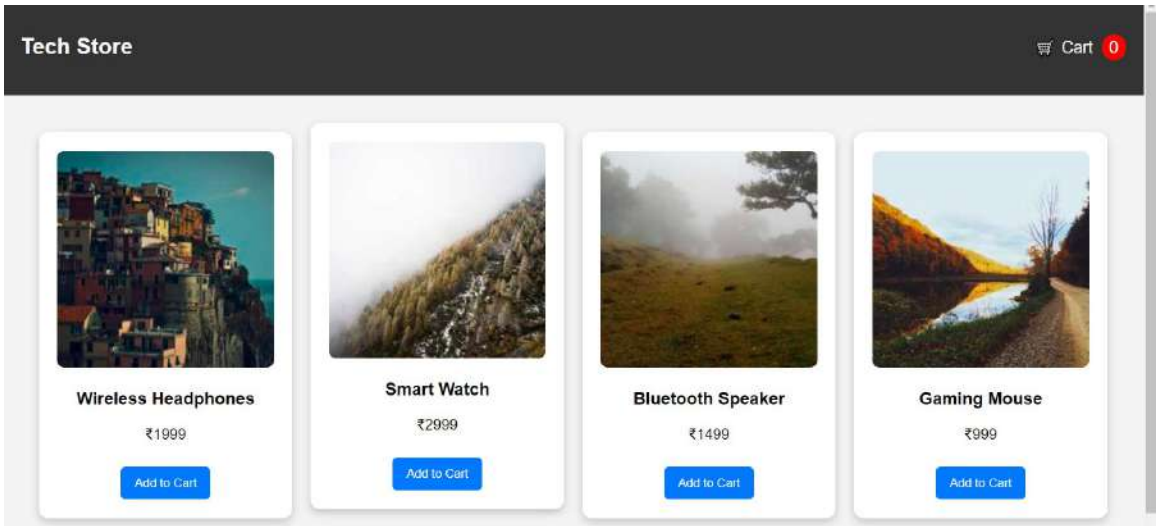
*Widely available across major browsers (Baseline since 2015)*

Syntax: <length-percentage [0,∞]>{1,4} [ / <length-percentage [0,∞]>{1,4} ]?

MDN Reference

JS script.js > ...

```javascript
var count = 0;
function addToCart(button){
count++;
document.getElementById("cart-count").innerText = count;
/* Visual feedback */
button.innerText = "Added ✓";
button.classList.add("added");
setTimeout(function(){
button.innerText = "Add to Cart";
button.classList.remove("added");
},1000);
}
```




**Explanation:**
AI tools such as GitHub Copilot were used to generate the initial structure for the e-commerce product page. The HTML code creates a grid layout that displays multiple products with images, names, and prices. CSS was applied to design the product cards and arrange them in a responsive grid layout to improve readability and user experience. JavaScript was implemented to add simple cart functionality where clicking the "Add to Cart" button increases the cart counter and provides visual feedback to the user.

**Task Description-5**
Create a Responsive Web Page Layout.
**Expected Output:**
A responsive web page with:
Header with navigation links
Main section with placeholder text/images
Footer with copyright or contact info

Layout adapts correctly to desktop, tablet, and mobile screen sizes.

**Prompt:**

Generate a responsive web page layout using HTML and CSS. Include a header with navigation links, a main section with placeholder images and text cards, and a footer with copyright information. Use responsive CSS with media queries so the layout adapts to desktop, tablet, and mobile screen sizes.

**CODE AND OUTPUT :**

INDEX.HTML > html

```html
<!DOCTYPE html>
<html>
<head>
<title>Responsive Web Layout</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<header>
<h2 class="logo">My Website</h2>
<nav>
<a href="#">Home</a>
<a href="#">About</a>
<a href="#">Services</a>
<a href="#">Contact</a>
</nav>
</header>
<main>
<section class="content">
<div class="card">
<img src="https://picsum.photos/300/200?1">
<h3>Web Development</h3>
<p>
Building responsive and modern websites using HTML, CSS, and JavaScri
</p>
</div>
<div class="card">
<img src="https://picsum.photos/300/200?2">
<h3>UI Design</h3>
<p>
Creating clean and user friendly interface designs for better
```

INDEX.HTML > html

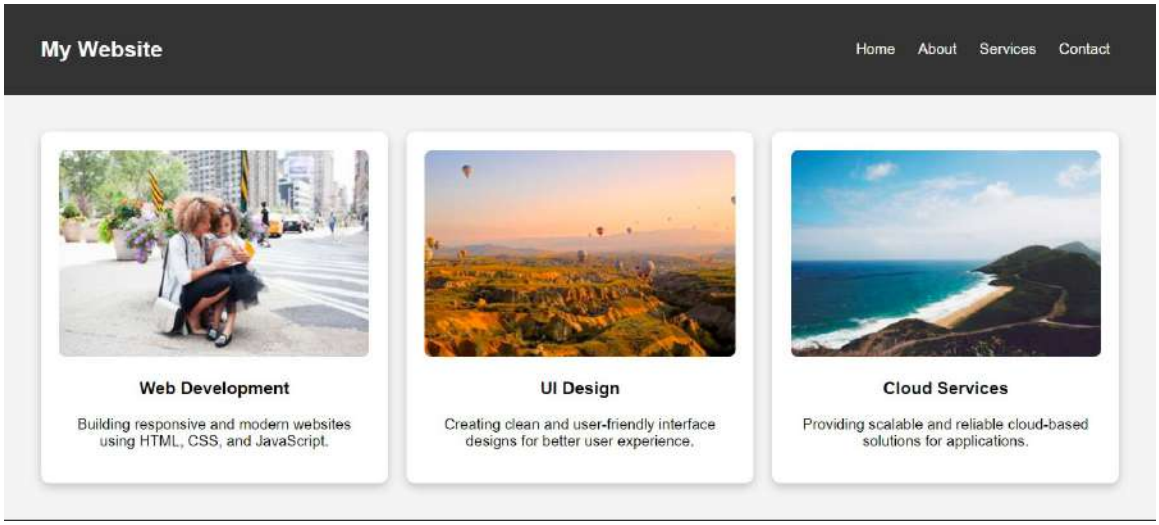
```html
<html>
<body>
<main>
<section class="content">
<div class="card">
Building responsive and modern websites using HTML, CSS, and JavaScri
</p>
</div>
<div class="card">
<img src="https://picsum.photos/300/200?2">
<h3>UI Design</h3>
<p>
Creating clean and user-friendly interface designs for better user ex
</p>
</div>
<div class="card">
<img src="https://picsum.photos/300/200?3">
<h3>Cloud Services</h3>
<p>
Providing scalable and reliable cloud-based solutions for application
</p>
</div>
</section>
</main>
<footer>
<p>© 2026 My Website | Contact: info@mywebsite.com</p>
</footer>
</body>
</html>
```

style.css > .content

```
body{
margin:0;
font-family:Arial;
background:#f4f4f4;
}
/* Header */
header{
display:flex;
justify-content:space-between;
align-items:center;
background:#333;
color:white;
padding:15px 40px;
}
nav a{
color:white;
text-decoration:none;
margin:0 10px;
font-size:16px;
}
nav a:hover{
color:orange;
}
/* Main Content */
.content{
display:grid;
grid-template-columns:repeat(3,1fr);
gap:20px;
padding:40px;
}
```

style.css > .content

```
25  .content{
30  }
31  .card{
32  background: white;
33  padding:20px;
34  border-radius:10px;
35  box-shadow:0 4px 10px rgba(0,0,0,0.2);
36  text-align:center;
37  }
38  .card img{
39  width:100%;
40  border-radius:8px;
41  }
42  /* Footer */
43  footer{
44  background:333;
45  color: white;
46  text-align:center;
47  padding:15px;
48  }
49  /* Tablet Responsive */
50  @media(max-width:900px){
51  .content{
52  grid-template-columns:repeat(2,1fr);
53  }
54  }
55  /* Mobile Responsive */
56  @media(max-width:600px){
57  header{
58  flex-direction:column;
```




**Explanation**:
AI assistance was used to generate a basic responsive webpage layout using HTML and CSS. The page structure includes a header with navigation links, a main content section containing cards with images and text, and a footer with contact information. CSS styling was applied to create a clean and visually organized layout using grid-based design. Media queries were added to ensure the layout automatically adjusts for different screen sizes such as desktop, tablet, and mobile devices.